

BLSN Report BLSN Synthetic Validation Report Theme: light Vehicle P3 Orion  
Cooling Mode helium\_closed\_loop Nominal Mass Flow 110 g/s Limits 80 - 140 g/s  
Sanity Check Telemetry Status Message PASS [PASS] PASS | Nominal mass flow is  
bounded by configured limits Tuple Execution Telemetry Targets (seconds) Metric  
Target Warn Threshold Fail Threshold ci\_total\_runtime\_s 120.0 180.0 240.0  
pytest\_runtime\_s 15.0 25.0 35.0 pipeline\_runtime\_s 5.0 10.0 15.0  
yaml\_json\_lint\_runtime\_s 3.0 6.0 9.0 Metric Markdown Slugs Ci Total Runtime S  
Pytest Runtime S Pipeline Runtime S Yaml Json Lint Runtime S Ci Total Runtime S  
Metric ID: `ci\_total\_runtime\_s` Target Threshold: **\*\*120.0s\*\*** Warn Threshold:  
**\*\*180.0s\*\*** Fail Threshold: **\*\*240.0s\*\*** Pytest Runtime S Metric ID:  
`pytest\_runtime\_s` Target Threshold: **\*\*15.0s\*\*** Warn Threshold: **\*\*25.0s\*\*** Fail  
Threshold: **\*\*35.0s\*\*** Pipeline Runtime S Metric ID: `pipeline\_runtime\_s` Target  
Threshold: **\*\*5.0s\*\*** Warn Threshold: **\*\*10.0s\*\*** Fail Threshold: **\*\*15.0s\*\*** Yaml  
Json Lint Runtime S Metric ID: `yaml\_json\_lint\_runtime\_s` Target Threshold:  
**\*\*3.0s\*\*** Warn Threshold: **\*\*6.0s\*\*** Fail Threshold: **\*\*9.0s\*\*** System Verification:  
Claimed vs. Actual Claimed Limits: 0 | Actually Tested Limits: 0 -&gt; STATUS:  
UNKNOWN Execution Velocity Total Runtime (engine + tests): 0.0000s Validation  
Strictness Pass: 0.00% | Fail: 0.00% | Skip: 0.00% Wave 12: SPC & Telemetry  
Stats Runtime Drift Current Runtime: 0.0000s Previous Moving Average: 0.0000s  
Historical Control Total Accumulated Runs: 1 Success Rate: 100.00% DoE Factors -  
A: python\_version , B: os\_type , C: load\_parameter Principal Component  
Eigenvalue Explained Variance Highest Weight Variable PC1 0.000000 0.00%  
total\_tests PC2 0.000000 0.00% skip\_rate Interactive Plotly Visualizations  
System Priority & Bottleneck Ranking (Bradley-Terry) Pairwise probability  
model:  $P(i \text{ \> } j) = p_i / (p_i + p_j)$  Global Repo Ranking Rank Artifact BT  
Strength Win Prob vs Top No global BT ranking data. Inter-Category Ranking Rank  
Category W  
in Prob vs Top No inter-category BT ranking data. QPLANT Requirements Ranking  
Bridge Rank Requirement Win Prob vs Top No QPLANT requirement ranking data.  
Mermaid System Flow Diagram graph LR; SSOT[SSOT Config] --> PIPELINE[Pipeline  
Execution]; PIPELINE --> MARKDOWN[Markdown Slug Pages]; PIPELINE --> INDEX[Index  
HTML]; PIPELINE --> SLIDES[Slides HTML]; INDEX --> PDF[PDF Export]; (function ()  
{ const emptyLayout = { xaxis: { visible: false }, yaxis: { visible: false },  
annotations: [{ text: 'No Plotly payload available', showarrow: false, xref:

```
'paper', yref: 'paper', x: 0.5, y: 0.5 }}]; const renderFallback = () => {  
Plotly.newPlot('plotly-telemetry-chart', [], emptyLayout);  
Plotly.newPlot('plotly-pca-chart', [], emptyLayout); };  
fetch('visuals/plotly_payloads.json') .then((response) => response.ok ?  
response.json() : Promise.reject(new Error('Missing Plotly payload')))  
.then((payload) => { const charts = payload && payload.charts ? payload.charts :  
{ }; const telemetry = charts.telemetry_combo || { }; const pca =  
charts.pca_bottlenecks || { }; Plotly.newPlot('plotly-telemetry-chart',  
telemetry.data || [], telemetry.layout || emptyLayout); Plotly.newPlot('plotly-  
pca-chart', pca.data || [], pca.layout || emptyLayout); }) .catch(() =>  
renderFallback()); })());
```